# Manual for updating the atrial fibrillation rhythm control treatment sequence model

Version 1.0
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# Table of contents

# 1 Introduction

### 1.1 Purpose of the manual
This document serves as support for adaptations to other countries or updates of the treatment sequence model for rhythm control in patients with atrial fibrillation. **Please read the technical report or scientific publication of the model before starting your update of the model.**

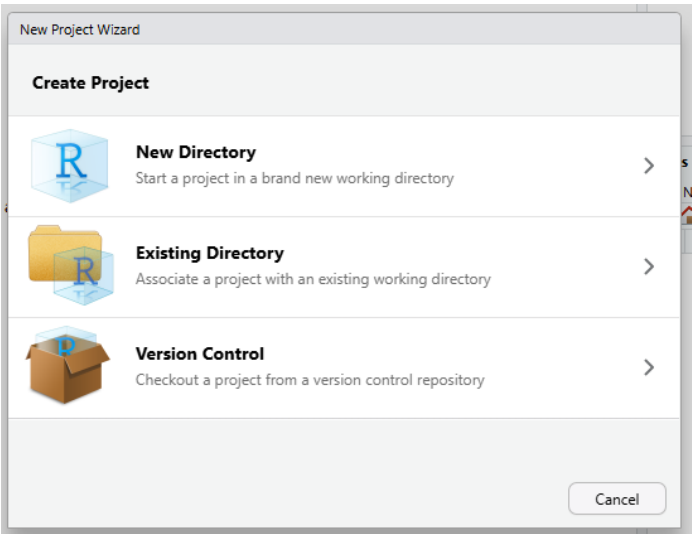
### 1.2 Collaborators
The treatment sequence model for rhythm control in patients with atrial fibrillation was developed in 2024-2025 in a collaboration with Zorginstituut Nederland, Cochrane Netherlands (for the systematic literature review and meta-analysis) and the 'Nederlanse Vereniging voor Cardiologie (NVVC)'. Find an overview of the collaborators in the table below.

| **Organisation** | **Name of involved staff** |
|---|---|
| Zorginstituut Nederland | Matthijs Versteegh<br>Simone Huygens<br>Marijke Delsing |
| Cochrane Nederland (Julius) | Demy Idema<br>Anneke Damen<br>Kevin Jenniskens<br>Lotty Hooft |
| NVVC | Martin Hemels<br>Dominik Linz<br>Michiel Rienstra<br>Sing-Chien Yap |

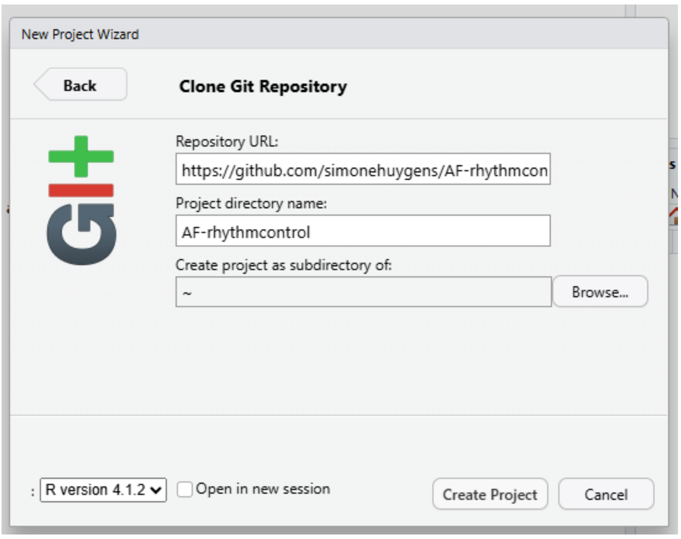
### 1.3 Prerequisites and required software/tools
Adapting or updating the model code requires access to and experience with R and R Studio and Git/Github.

### 1.4 Cloning the model into R from Git
Please follow the following steps to load the model in to the R Studio environment.

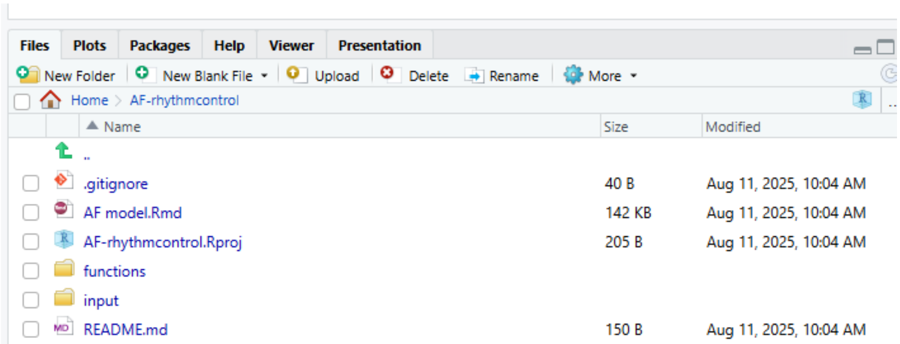
1. Open RStudio
   *For Zorginstituut:* Open the Posit workbench via the Microsoft Edge browser and select '+ New Session'.
2. Click on 'File' in the top left and then on 'New project' to open a new project.
3. You will see the menu below, select 'Version Control' and then 'Git'.



4. Fill in the Git repository link of the AF model at 'Repository URL': [https://github.com/simonehuygens/AF-rhythmcontrol.git](https://github.com/simonehuygens/AF-rhythmcontrol.git)
   It is a public git repository, so no password is needed.
5. The 'Project directory name' will then be filled in automatically.




6. Click on 'Create Project' and the project and associated files will be loaded.
7. In the bottom right (usually, but depending on your own RStudio layout) under the 'Files' tab you will now see the files for the AF model. Open the 'AF model.Rmd' file.

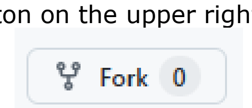



The 'input' folder contains all necessary input files. The 'functions' folder contains the 'functions.R' script that is needed to run the model and is automatically loaded when running section 1.3 of the 'AF model.Rmd'.

### 1.5 Log code changes or 'fork' the model

Please make a log of the code changes that have been made. Ideally, start a new repository on Github with the old code as source (using the fork option on GitHub) and the updates as new commits so that version control is possible all the way back to the original model. This is an alternative to cloning (described in 1.4).

To do this you can click the Fork button on the upper right of the [repository's GitHub page](https://github.com/simonehuygens/AF-rhythmcontrol).




GitHub will create a complete copy of the model under your own account with your own username/repository-name version. You can make changes, commit and push to your fork freely.

# 2 Adapt or update input parameters

### 2.1 Settings

Section 1.2 describes the main settings of the model. Most settings should not be adapted. Only the following settings can be changed or adapted as needed:

- The minimum age (min_age) should not be lower than 18, because the model is developed for adults.
- The maximum age (max_age) should be equal to the maximum age in the background mortality table.
- The number of patients (n_i) should be at least 50,000 for stable model results, but for practice run this can be reduced to increase the speed of the model simulations.
- For country adaptations, the willingness to pay threshold (wtp), discount rate for costs (d_rc) and discount rate for effects (d_re) can be adapted.

```
# Model settings
min_age    <- 18   # min age: only include adults
max_age    <- 108  # max age based on max in Human Mortality Database
…
n_i        <- 50000 # number of simulated individuals
…
wtp        <- 20000 # willingness to pay threshold
d_rc       <- 0.030 # discount rate costs
d_re       <- 0.015 # discount rate effects
```

### 2.2 Patient population

The characteristics of the patient population are set in section 2.
Users can adapt the mean and standard deviation of age, the proportion of females and the proportion of paroxysmal versus persistent AF. Based on these inputs, the model will generate a patient population with varying ages and sex.

When changing the age, the calculation of the weighted average of age of survived and deceased patients can be removed, as this was only included for the specific source (de Mol et al. 2022) that was used for age in the original model.

```
#### PATIENT CHARACTERISTICS ####
n_survived           <- 30182  # de Mol et al. 2022 (NHR 2013-2020) Table 1
n_deceased           <- 15     # de Mol et al. 2022 (NHR 2013-2020) Table 1
n_total              <- n_survived+n_deceased
age                  <- ((61.4*n_survived)+(65.9*n_deceased))/n_total
sd_age               <- ((9.8*n_survived)+(7.1*n_deceased))/n_total
n_female             <- 9816+7
n_male               <- n_total-n_female
prop_female          <- n_female/(n_female+n_male)
n_paroxysmal         <- 19527+11      # proportion paroxysmal AF
n_persistent         <- 7300+3+542+0 # proportion persistent AF
prop_parox           <- n_paroxysmal/(n_paroxysmal+n_persistent)

# Generate patient population (will be varied in PSA)
set.seed(2)
v_age                <- round(rtruncnorm(n_i, min_age, max_age, age, sd_age))
v_Sex                <- rbinom(n_i, 1, prop_female)
```

### 2.3 Clinical input parameters

The model consists of the following clinical input parameters:

| Input parameter | Source |
|---|---|
| Relative risks of AF recurrence AAD vs. CA | Meta-analysis Cochrane Nederland |
| Probability of recurrence of AF after CA after 1 year | Meta-analysis Cochrane Nederland and health insurance claims data from Vektis |
| Long-term recurrence of AF symptoms after CA | Health insurance claims data from Vektis |
| Background mortality | Human Mortality Database |
| Excess mortality | Vinter et al. 2020 |
| Discontinuation of AAD's due to adverse events | Meta-analysis Cochrane Nederland |
| Adverse events CA | NHR ablation analysis tool (Behandelgroep: Katheterablatie AF, year: 2023) |

This chapter explains how these input parameters can be adapted.

*2.3.1 Relative risks and short term AF recurrence*

The relative risks (RR_nai, RR_AAD_exp_parox, RR_AAD_exp_pers, RR_CA_exp*) and probabilities of recurrence after 1 year (prop_AF_CA_nai, prop_AF_CA_AAD_exp_parox, prop_AF_CA_AAD_exp_pers, prop_AF_CA_CA_exp_1*, prop_AF_CA_CA_exp_2*) are mainly based on the meta-analysis. The results of the meta-analysis are loaded in section 2 of the .Rmd. This loads the data that is needed to calculate the relative risks of AAD versus CA and the probabilities of recurrence of AF 1-year after the CA (see code below).

```
#### Probabilities of treatment success ####
# Load results of meta-analysis Cochrane Nederland
load(here::here("input", "meta-analysis-efficacy.RData"))
load(here::here("input", "meta-analyse disc_AAD.RData"))
```

To update these RData objects, the original scripts of the meta-analysis need to be updated. However, this script and underlying data extraction file is not publicly available. Users can manually adapt the parameters if they are provided in other sources (for example a publication of a new meta-analysis) or contact Cochrane Netherlands to update the meta-analysis with the same methods as the original meta-analysis. Please be aware that for the OWSA you ideally need the lower and upper limit of the 95% confidence intervals and for the PSA you need the standard error of the input parameters.

*Currently these input parameters were not available in the meta-analysis, therefore they were derived from health insurance claims data.

*2.3.2 Long-term AF recurrence*

These input parameters are based on the health insurance claims data that is loaded from the input folder using the following code.

```
load(here::here("input", "TTE_AF_Vektis.RData")) # survival model objects
(version 17-04-2025)
```

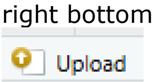
If an update of the data is available, this RData object needs to be updated. This RData object consists of three flexsurvreg objects called best_fit_ablation 1 to 3. This contains the details of the best fitting parametric distribution of the Kaplan-Meier curve of the first, second and third or higher CA estimated on the health insurance claims data.

These can be replaced with flexsurvreg objects that are estimated on updated data or other data sources.

*For Zorginstituut:*

If an update of the model is required, contact Marijke Delsing to update the data file generated on 17 April 2025. Then follow the following steps to update the flexsurvreg objects:

1. Save the 'alle' tab in a separate .csv file.
2. Save the .csv file in the 'input' folder of the R project using the upload button in the right bottom panel of R studio.

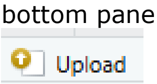
3. Open the R script 'Update Vektis time to AF recurrence analysis.R' in the 'functions' folder of the R project.
4. Change the file name of the .csv file in the R script to the new file name.
5. Run the R script 'Update Vektis time to AF recurrence analysis.R' with the updated .csv file. This will automatically save the best_fit_ablation objects in the 'input' folder under the name that is specified at the bottom of the R script.
6. Change the file name of the .RData file to the new file name.

```
if(scenario == "basecase"){
  load(here::here("input", "TTE_AF_Vektis.RData")) # survival model
objects (version 17-04-2025)
```

*2.3.3 Mortality*

To update or change the source for background mortality in the general population create a .csv file with a column for age ("Age") and columns for mortality rates for females ("Female") and males ("Male").

Save the .csv file in the 'input' folder of the R project using the upload button in the right bottom panel of R studio.




Change the reference to the .csv file in the code in section 2 of the .Rmd (see below). Note that you may have to change the separator in read.csv. Additionally, the mortality rate of the highest age in the mortality table should be set to 1. The code to do this is already provided (see the bottom of the code below), but you need to make sure 108 is the maximum age and if not adapt it to the maximum age in your mortality table.

```
#### Probability of death ####
# Background mortality in the general population weighted by the sex
distribution of the patient population
# We used 2019 to avoid bias from COVID-19
df_mort              <- read.csv(here::here("input", "sterfte_NL_2019.csv"),
sep = ";", header = TRUE)
df_mort              <- df_mort[ , -c(1,5)] # remove year and total column
df_mort <- df_mort %>% # convert table from wide to long with a variable for
sex
  pivot_longer(
    cols = c(Female, Male),
    names_to = "Sex",
    values_to = "r_mort"
  ) %>%
  mutate(
    Sex = ifelse(Sex == "Female", 1, 0),
    r_mort = as.numeric(r_mort)
  )
df_mort$r_mort_cl  <- df_mort$r_mort*cl # convert annual mortality rate to x-
weekly mortality rate
df_mort$p_mort_cl  <- 1-exp(-df_mort$r_mort_cl)     # convert x-weekly
mortality rate to probability
df_mort[df_mort$Age == 108, "p_mort_cl"] <- 1 # set mortality rate to 1 at max
age to make sure everyone dies in the model
```

The hazard ratio for excess mortality can be changed here:
```
HR_EM <- 2.01 # 95% CI: 1.71 to 2.36 (Vinter et al. 2020)
```

*2.3.4 Adverse events*
For AADs adverse events that result in discontinuation of AADs are included in the model. The probability to discontinue is derived from the meta-analysis and can be updated in section 2 (see below).

```
#### Probability of severe adverse events ####
# Discontinuation of AAD's due to adverse events
p_disc_AAD_nai <- 0.121 # Source: meta-analysis Cochrane Netherlands
p_disc_AAD_exp <- 0.225 # Source: meta-analysis Cochrane Netherlands
```

For CA three complications: cardiac tamponade, phrenicus paralysis and vascular complications are included in the model. The number of events and total number of CAs can be updated in section 2. If these are not available the probabilities can directly be imported in the p_CT_CA, p_PP_CA and p_VC_CA variables.

```
# Adverse events CA (Source: event rates from NHR ablation analysis tool:
Behandelgroep: Katheterablatie AF, Jaar: 2023)
n_CT_CA         <- 26   # cardiac tamponade
n_tot_CT_CA     <- 6150
p_CT_CA         <- n_CT_CA/n_tot_CT_CA
n_PP_CA         <- 25   # phrenicus paralysis
n_tot_PP_CA     <- 6211
p_PP_CA         <- n_PP_CA/n_tot_PP_CA
n_VC_CA         <- 40   # vascular complications
n_tot_VC_CA     <- 6149
p_VC_CA         <- n_VC_CA/n_tot_VC_CA
```

**2.4 Costs**
Healthcare costs are loaded in df_c_states, see below:
```
#### Health state costs and costs of interventions (Source: Vektis)
df_c_states  <- load(here::here("input", "c_Vektis.RData"))
```

This loads the following input parameters:

| Data | |
|---|---|
| c_AAD_params | List of 17 |
| c_CA_DBC_params | List of 17 |
| **Values** | |
| c_AAD | 51.289397993311 |
| c_CA_DBC | 11105.5982207358 |
| c_d_after_CA1 | 643.063444816054 |
| c_d_after_CA2 | 93.5856722408027 |
| c_d_before_CA | 739.286528428094 |
| c_d_SAF | 830.249444816053 |
| c_sd_d_after_CA1 | 6163.50101959616 |
| c_sd_d_after_CA2 | 4500.5095504739 |
| c_sd_d_before_CA | 4392.83348669121 |
| c_sd_d_SAF | 4476.6291761755 |
| df_c_states | chr [1:13] "c_d_SA |
| n_c_Vektis | 1495L |

These input parameters and associated standard deviations can be changed manually if other data sources are available. Please be aware that variables that start with c_d are cost differences compared to no AF symptoms.

*For Zorginstituut:*
If an update of the model is required, contact a business analyst (e.g. Marijke Delsing) to update the data file. Then follow the following steps to update input parameters:

1. Save the .xlsx file in a .csv file.
2. Save the .csv file in the ‘input’ folder of the R project.
3. Open the R script ‘Update Vektis cost analysis.R’ in the ‘functions’ folder of the R project.
4. Run this script with the updated .csv file and save the results in an .RData file in the ‘input’ folder (the code to do this is already provided ad the end of the script).

Societal cost calculations can be changed in the code if new data becomes available.

**2.5 Correct costs for inflation**
The costs are corrected for inflation with the adjust_inflation function in the ‘functions.R’ script. If the costs need be updated with a new correction for inflation, the target_year and inflation rate for future years can be changed in the ‘functions.R’ script. The target_year should be changed to the current year. The infl_rates for the following year can be added under 2023. NB. The inflation rate for 2023 is actually the rate that is reported at 2024 at ‘Jaarmutatie CPI afgeleid (%)’ in CBS (but it is the inflation rate from 2023 to 2024 and is therefore reported with “2023” here).

```
#### Adjustment for inflation using consumer price index ####
# Cost year = 2024, except for drug costs (2025)
# Correct for inflation
adjust_inflation <- function(amount, source_year, target_year = 2024) {
  # Define yearly inflation rates (as decimals)
  infl_rates <- c(
    "2017" = 0.014,
    "2018" = 0.016,
    "2019" = 0.012,
    "2020" = 0.025,
    "2021" = 0.118,
    "2022" = 0.030,
    "2023" = 0.024
  )

  # Get years to include in calculation
  years <- as.character(source_year:(target_year-1))

  # Calculate inflation factor and apply
  factor <- prod(1 + infl_rates[years])
  return(amount * factor)
}
```

**2.6 Utilities**
The utility input parameters consists of three elements: the disutility of having AF symptoms (du_SAF), the disutility for complications after CA (du_CT_mo, du_PP_mo, du_VC_mo), and the utility without AF symptoms which is assumed to be equal to the general population utility (mod_splines).

The disutility of having AF symptoms or complications after CA can be replaced with another value if there is another source available or preferred.

The general population mortality should be adapted to the local setting if possible. In the original version of the model it is included in a regression formula with age and sex as covariates. In the Effs function (in section 03.6) this model is used with the predict function to calculate the utility associated with the patients current age and sex. The mod_splines object can be replaced with another R object of a regression.

```
#### UTILITIES INPUTS ####
#### Disutilities AF and adverse events ####
# Load utilities based on the EQ-5D-5L data in the AVATAR-AF trial using the Dutch tariff
load(here::here("input", "glm_AVATARAF_NL.RData"))

du_SAF       <- abs(unname((glm_AVATARAF_NL$coefficients[2]))) # Re-analysis of AVATAR-AF with EQ-5D-5L NL tariff based on Moss et al.

# Assumption similar to Akerborg et al. (2012) and Reynolds et al. (2014)
du_CT_mo  <- 0.1 # Assumption disutility of 0.1 for 1 month, corrected to disutility for the cycle length in Effs
du_PP_mo  <- 0.1 # Assumption disutility of 0.1 for 1 month, corrected to disutility for the cycle length in Effs
du_VC_mo  <- 0.1 # Assumption disutility of 0.1 for 1 month, corrected to disutility for the cycle length in Effs

#### General population utilities ####
# Regression model general population by age and sex (based on Versteegh et al. 2016)
load(here::here("input", "splines_HRQoL.RData"))
gen_pop_utility  <- "fit"
mod_splines_coef <- mod_splines$coefficients
```

## 3 Functions

The model uses several functions for the microsimulation:

- Create_df_X: creates a dataframe with patient characteristics (section 3.1).
- PrepareProbs: calculates all required clinical input parameters for the model from the clinical input parameters defined in section 2 (section 3.2).
- Probs: creates the transition probability matrix for each patient (section 3.3) and is based on the DARTH modelling framework.[refs]
- PrepareCosts: calculates all required cost input parameters for the model from the cost input parameters defined in section 2 and adjusts the inputs for inflation if necessary (section 3.4).
- Costs: calculates the costs per cycle for each patient (section 3.5) and is based on the DARTH modelling framework.[refs]
- Effs: calculates the QALYs per cycle for each patient (section 3.6) and is based on the DARTH modelling framework.[refs]
- MicroSim: combines all functions to run the microsimulation (section 3.7) and is based on the DARTH modelling framework.[refs]

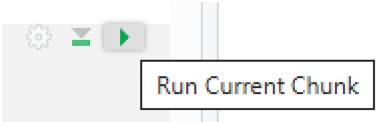
For more information on the individual functions we refer to the annotations in the .Rmd and/or the DARTH modelling framework literature.[refs]

## 4 Run analyses

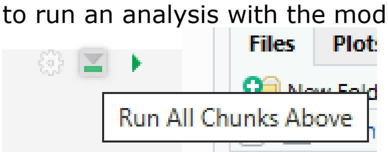
After the model is adapted, the model can be used to run analyses.

For basic users, we advise to use the ‘AF model.Rmd’ to run the model.
When using the ‘AF model.Rmd’, we advise to either:

- run the chunks one by one manually by clicking on ‘Run Current Chunk’




- or to go to section 04.1 and click on 'Run All Chunks Above' to load the complete model and the input parameters. Then run 04.1 or any of the other chunks below to run an analysis with the model.




- It is not recommended to run all chunks from top to bottom of the .Rmd consecutively because this includes sensitivity analyses that have a long runtime and there is a higher likelihood of mistakes.

In 4.1 the analysis can be run for one sequence. The treatments TRT1 to TRT6 can be adapted by the user and should either be "CA" or "AAD".

```
outcomes <- MicroSim(l_params, n_i, df_X, TRT1 = "CA", TRT2 = "AAD", TRT3 =
"AAD", TRT4 = "AAD", TRT5 = "AAD", TRT6 = "CA", seed = 1)
```

There are several analyses available in the .Rmd:

- Deterministic analysis of 1 sequence (section 4.1)
- Deterministic analyses of all sequences (section 4.2), including health state and line traces and summary tables.
- One-way sensitivity analyses (section 5), including tornado diagrams of costs, effects and the net health benefit.
- Probabilistic sensitivity analyses (section 6), including creating cost-effectiveness planes, cost-effectiveness acceptability curves, and value of information analysis.
- Model stability testing (section 7) which can estimate the number of individuals (n_i) that should be simulated to have stable deterministic results.

In addition to the .Rmd, the 'functions' folder contains scripts to run a base-case analysis, scenario analysis, one-way sensitivity analysis or probabilistic analysis and to test the model stability. These scripts can be useful when you want to start a 'Workbench job' in the Posit Workbench which enables you to run several analyses at the same time while still being able to use R for other analyses. Please make sure to adapt the 'model functions.R' script (or any of the other scripts) with the changes you made in the 'AF model.Rmd' in case you want to use any of these scripts to run analysis.